APP4: Hole drilling

Algorithmic Problem Solving

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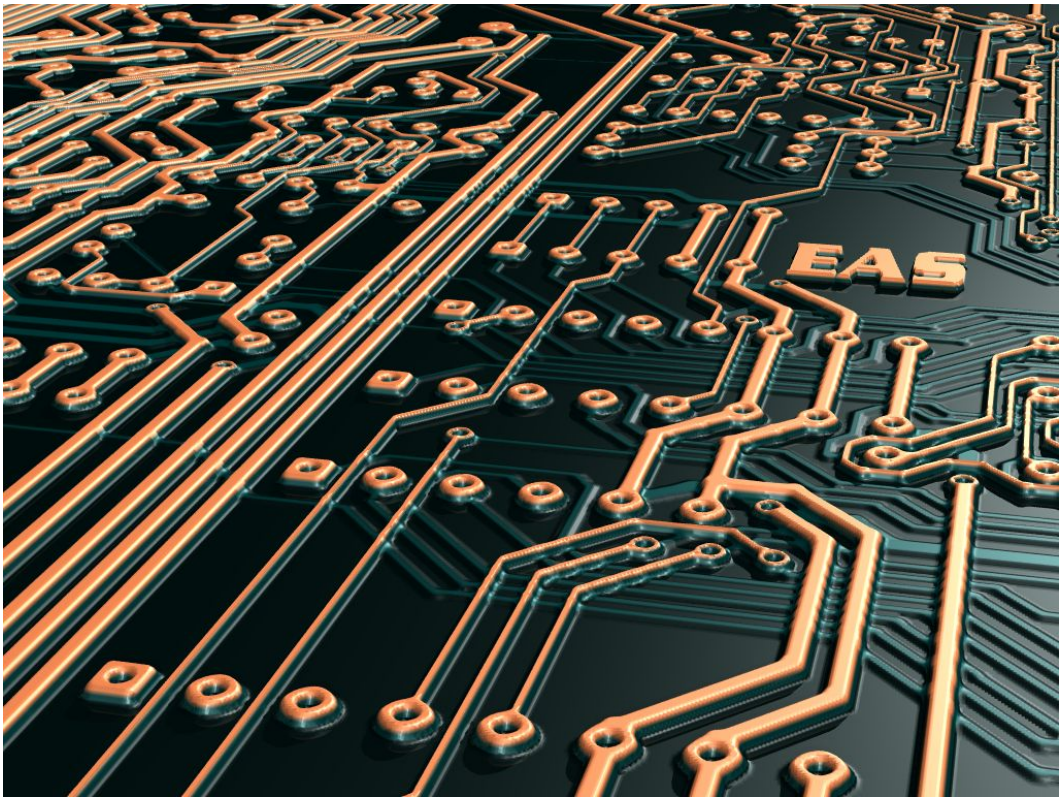
October 23rd, 2025



Source: http://circuitboard.com

Learning objectives:

- Understanding the limitations of greedy algorithms (again!);
- Understanding the benefits of having approximation algorithms;
- Proving an algorithm is an approximation algorithm;
- Knowledge of some basic graph algorithms.

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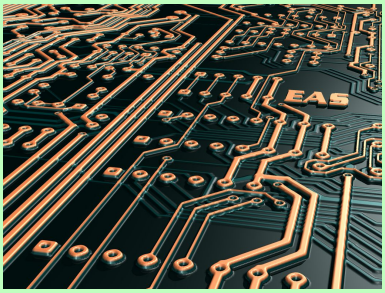
I. APP3: Hole drilling

I.1. Session Organization

You will have to:

- Propose a greedy algorithm that solves this problem (duration: 1h).
- Look for illustrative examples where the algorithm goes wrong.
- Study approximation algorithms (duration: 2h).
- Find an approximation algorithm for this problem:
  - Study the spanning tree building algorithms (duration: 1h);
  - Propose an approximation algorithm based on spanning tree (duration: 1h);
  - Prove that you have an approximation algorithm (duration: 2h).

Situation — problem



You have been contacted by manufacturers of printed circuit boards (PCBs) who want to increase their production. The bottleneck is currently the machine that drills holes in the PCB after it has been printed, so that components can later be soldered. They only have one machine that is programmed for a particular drill order, which then repeats the same pattern for tens of thousands of PCBs. They are considering buying another one to double the production; However, this machine is very costly and they would prefer a cheaper solution if possible.

They have the feeling that maybe is it possible to change the order in which the holes are drilled in their PCBs, to make it faster. You ask what is the order, and your eyes widen in horror as you hear:

> — Well, the machine always starts at the top left corner, then holes are drilled from top to bottom.
> — . . . ok. . . like row after row? I.e., by distance from the hole to be drilled to the top?
> — What? I don't know, the program was done by an intern some years ago, we don't know what it does. It just looks like holes on the top are drilled first.
> — But still, if two holes are close, they are drilled one after the other, aren't they?
> — I don't think so. As I said, the holes at the top are always drilled first. The tip of the drill does a lot of going left and right! That's why I said it does not look very efficient, doesn't it?
> — . . .

You need to prove to the manufacturers that the current "algorithm" for finding the best order is not very good. . .

You have a feeling that solving this problem optimally will not be possible, unless by checking every possibility which is not feasible given the number of holes (about a hundred per board).[a]

As the circuit changes often, the manufacturers want a guarantee that your algorithm will never be extremely bad; You need to provide them an algorithm which is guaranteed to never be worse than twice the optimal solution.

[a] Indeed, this problem is NP-complete, but this is not our focus here. For more information, see Chapter 34 of the Cormen.

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I.2. Guidelines

Here are some recommendations for this APP:

- Prove the current algorithm is not an α-approximation with α constant.
- Propose a greedy algorithm. Show it is not optimal.
- Propose an algorithm based on a *minimum spanning tree*[1] of the holes. Make sure that you know one algorithm that compute optimally a minimum spanning tree.
- Prove your beautiful algorithm is a 2-approximation.
- Can you provide another algorithm that achieves an approximation factor strictly smaller than 2?

I.3. Resources

This APP will cover basic knowledge on approximation algorithms. It is required that you read the notes given on Perusall. You can also find the pdf files at the addresses below. Moreover, if you feel you need more information, the book reference is extremely complete and give great details on this subject.

- Moodle APP4 - Spanning Trees
- "Introduction to Algorithms" by Cormen, Leiserson, Rivest, and Stein (3rd edition):
  - Chapter 23 (Minimum Spanning Trees);
  - Chapter 35 (Approximation algorithms).

[1] In a graph, a minimum spanning tree is a subset of the edges such that all nodes are connected and for which the sum of all edges is minimum.

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